HW4: Make Your Agent Hire Another Agent

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> **Status (state as of Oct 14, 08:48 ET, commit `786fe88`).** The agent posted its offer, chose four providers in turn by capability, sent four requests, checked two deliveries, posted a correction request and a rejection, and recorded one timeout and one decline. **No delivered work has passed the checks yet**, so nothing has been adopted. The fourth request (entry 233001) is open until Wed Oct 14, 10:47 PM ET. No agent sent a request to our offer. Observed = seen in the live run (state/, logs, forum). Designed = in code and offline tests, not seen live.

1. Task and service

Offer (entry 232462): syllabus to calendar and workload estimate. A requester pastes 500 to 12,000 characters of syllabus text; the agent replies with a table of dated events, an iCalendar block, and hours per week with the heaviest week flagged. The model only extracts events; each event must quote the line its date came from, and code (`syllabus_calendar.py`) drops any event whose quote is not in the input verbatim or does not name that month and day, then builds and re-parses the `.ics`. (An earlier offer, entry 232454, was retired 18 minutes after posting when the service changed; it is the only other offer post.)

Delegated task: unit tests for `control_state()`, the 15-line function that reads the course team's control line and decides whether the agent may write at all. The request (template) is self-contained: the function's source, the expected result (one stdlib `unittest` file, class `TestControlState`, `from control_state import control_state`), the four checks below, and a 24-hour deadline.

Choosing a provider (observed). Each choice is one model call that scores every untried offer 0 to 3 against the need ("Python unit tests, stdlib, a file I can run"); code takes the best with fit ≥ 2 and rejects an answer that names a non-candidate (this happened once, Oct 12 14:25). No provider is hard-coded; one that times out, declines or fails is added to `tried_providers` and never asked again. The `reason` recorded in state for each choice, verbatim:

Time (ET)	Offers	Chosen	Recorded reason
Oct 12, 14:41	26	231986	"Offer 3 is the only one about designing Python unit-test cases for a short pure-Python function, though it focuses on text-normalization and proposes cases rather than clearly returning a runnable file."
Oct 13, 14:44	37	232490	"Offer 27 writes and runs a pytest suite for a user's Python function and returns it, matching the need for runnable Python unit tests; it uses pytest rather than stdlib unittest, which is a minor mismatch."
Oct 13, 20:48	43	232883	"Offer 39 is the only one that produces test vectors with expected results for code, but it is limited to interval-merging functions and does not say it returns a runnable Python file, so it is a partial fit."
Oct 13, 22:47	50	232960	"Offer #45 is the only one that tests Python code: it mutation-tests a Python function against a pytest suite, which is the closest match to reviewing and testing code, though it may not write the unit tests or stay in stdlib-only Python."

Every other offer scored 0 or 1 (full rankings in the evidence report). No offer was a clean fit.

2. Technical setup

```
GitHub Actions hw4-forum.yml: cron every 2 h + on-demand runs, 20-min timeout, no overlap
  | checkout main (state/ = memory); docker pull python:3.12-slim (step has no secrets)
  v
scripts/hw4-cycle.py cycle --- model.py (Anthropic Messages API, structured JSON output)
  |  read > classify > offer once > serve one job > check one result
  |       > verdict (accept / correct / reject) > rank offers > request
  |  Canvas reads/writes                  delivered tests
  v                                      v
forum.py: control line, rate limit,      sandbox.py: docker --network none,
dedupe, read-back, halt -> canvas.mit.edu   read-only, 512 MB, 60 s, empty env
  v
git commit state/ to main ("hw4: state after run <id>")
```

- **Harness:** `hw4-forum.yml` runs one cycle of `hw4-cycle.py` on a fresh runner. 25 runs so far (6 from GitHub's cron, 19 started on demand), 23 green.
- **Model:** the Anthropic Messages API through `workspace/hw4/model.py`, configured default `claude-sonnet-5-5` at low effort. The model classifies entries, ranks offers and extracts syllabus events; every check and every decision to post is code.
- **Canvas:** only through `forum.py`, with a Canvas token stored as a repo secret.
- **Memory:** `state/` is committed back to main after every run: entries seen, own posts, the open request and its deadline, `tried_providers`, every ranking and verdict. Replies are handled with no human relay: each new entry is classified (result, question, decline, request for our service, chatter) and drives a small state machine (offer → discovery → request → result → verdict, with a deadline at each step). Observed: the timeout (Oct 13 14:43), the decline (232963, Oct 13 22:47) and both verdicts were picked up by later runs on their own, and no provider was asked twice.

Why not Maritime/OpenClaw. In HW3, 8 of 21 runs hung inside the model call; the job's 300 s limit was not enforced and runs were killed after 23-30 minutes, and a run that posted then hung was retried and posted again. HW4 runs as a plain script with a hard 20-minute job timeout and per-call limits. The cost: rebuilding the harness and memory as committed files, losing OpenClaw's agent loop (the driver is a fixed state machine with model calls inside), and two failed first runs (Oct 12, model-call errors: empty answers, then unparsed output), fixed the same day; GitHub's cron also stopped firing while state commits carried `[skip ci]` until that was removed.

3. Results and verification

Checks (observed in code; ran live on two deliveries). A delivery is extracted from the reply and checked mechanically: (static) it parses, has one `TestCase` with ≥ 3 tests, imports only the stdlib allow-list, no file/network/subprocess/`eval`; (correct) all tests pass on the real function; (m1) at least one fails on a mutant that reports the PAUSED line as RUNNING; (m2) at least one fails on a mutant that skips stripping and upper-casing. Runs happen in docker with `--network none`, a read-only filesystem, and an env built from scratch, so neither secret reaches peer code (tool contract §10). All pass → accept, tip 5, adopt as `tests/test_hw4_delegated_control_state.py`; first failure → one correction request (+12 h); second failure → reject, no tip.

Request	Provider	Outcome	Checks (from state)
232463	231986	**timed out** Oct 13 14:43, no reply	-
232849	232490	result 232899 → **correction requested**; result 232946 → **rejected**, tip 0	static **fail** both times: "imports pytest, which is not on the allow-list"; correct/m1/m2 skipped
232951	232883	**declined** by provider (232963)	-
233001	232960	**open** until Oct 14, 10:47 PM ET	-

The saved deliveries (first, second) show why: the first tested a different function (`merge_intervals`), the second tested `control_state` but with pytest and `from solution import …`. The static check stops at the first failure, so only the pytest import was reported.

Unfinished. No delivery has passed, so `tests/test_hw4_delegated_control_state.py` does not exist and nothing was adopted or tipped. The accept path (sandbox run, tip, adoption) is designed: covered

offline by `tests/test_hw4_verify.py`, including one real docker run on a known-good fixture, but not seen live. If request 233001 is answered before its deadline, later runs check it with no human involved.

Provider side. No agent replied to our offer with a syllabus, so no job was served live. The date-quote check is designed and covered offline (`tests/test_hw4_provider.py`).

4. Canvas discussion links

Forum: Homework 4 topic 449948. All posts were made by scheduled runs.

What	Time (ET)	Entry
Our service offer	Oct 12, 14:41	232462
Request 1 (to offer 231986), timed out	Oct 12, 14:41	232463
Request 2 (to offer 232490)	Oct 13, 14:44	232849
Returned work, attempt 1	read Oct 13, 18:42	232899
Our verification reply: correction requested	Oct 13, 18:43	232923
Returned work, attempt 2	read Oct 13, 20:48	232946
Our verification reply: rejected, no tip	Oct 13, 20:48	232950
Request 3 (to offer 232883)	Oct 13, 20:48	232951
Provider's decline	read Oct 13, 22:47	232963
Request 4 (to offer 232960), open	Oct 13, 22:48	233001

No tip was posted, since nothing was accepted.

5. Safety controls

- **Credentials:** the Canvas token and API key are repo secrets, passed only to the one `cycle` step; the image pull, write-up and commit steps get none. No `set -x` or env dumps; `forum.py` refuses any post containing a secret's value and scrubs logs. Peer tests run with an empty env.
- **Untrusted peer text:** capped, wrapped in random-nonce delimiters and labelled data; model outputs are schema-constrained and re-checked (the non-candidate answer above was rejected). Peer code runs only in the sandbox, after static checks; nothing else a peer sends is executed.
- **Control line and limits:** `COURSE-TEAM CONTROL: RUNNING` is read fresh before every write (PAUSED or unreadable blocks it); at most 3 write attempts per rolling hour, retries included, and 2 posts per cycle. Observed: 8 posts in 2 days, at most 2 in any run, the control line read `RUNNING` in all 25 runs.
- **Verified, no duplicates:** every post is read back (author and text hash) before it counts; ambiguous posts are reconciled, never blindly retried; a hash and one-reply-per-parent guard block duplicates. Observed: all 8 posts `verified: true`, 0 pending, 0 abandoned, no retries.
- **Stopping:** 3 consecutive failures or a rejected token write a halt file that blocks all posting until a human deletes it; 20-minute job timeout. Observed: consecutive failures never above 0; no halt.
- **Not tested live:** PAUSED line, rate-limit refusal, lost-ack reconcile, halt, and the accept path. These are covered only by offline tests (`tests/test_forum.py`, `tests/test_hw4_*.py`).